7th Grade Math · Weekly Check · Unit 10 C-AR.2.2

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

Learning goal: I can write and solve two-step equations. (circle your level after scoring)

4	I can write a two-step equation from a real-world context, solve it, and check the answer.
3	I can solve a two-step equation, including with rational numbers or the distributive property.
2	I can solve a one-step equation.
1	I need help to start this benchmark.

1. A park charges a $25.00 entry fee plus $12.50 for each hour of skate rental. A visitor paid $62.50 in all.

Write an equation that can be used to find h, the number of hours of skate rental the visitor paid for. Then solve the equation.

2. Solve the equation for h.

$1.5(h + 1.6) = -0.6$

Turn over →

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____

3. 3 students each saved the same amount, s dollars, and then each earned \$5.00 more. Together they now have \$105.00.

Write an equation that can be used to find s. Then solve the equation.

4. What is the exact value of g in the equation shown?

$$-1\frac{7}{9} = \left(\frac{2}{3}\right) g - 4$$

5. Gym A charges \$24.50 each month. Jordan used a coupon worth \$35.00 off the total and paid Gym A \$185.50 in all.

Write and solve an equation to find n, the number of months Jordan paid for. Gym B charges \$26.50 each month and takes no coupons. For the same number of months, which gym would have cost less, and by how much?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.2.2 · MA.7.AR.2.2 · Two-step equations with rational numbers

Unit 10: Equations and Inequalities. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $12.5h + 25 = 62.5$; $h = 3$. **Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

The cost of the hours of skate rental plus the entry fee equals the total paid: $12.5h + 25 = 62.5$.
Subtract 25 from both sides: $12.5h = 37.5$.
Divide both sides by 12.5: $h = 3$.

2. Regular $h = -2$

Divide both sides by 1.5: $h + 1.6 = -0.6 \div 1.5 = -0.4$.
Subtract 1.6 from both sides: $h = -0.4 - 1.6 = -2$.

3. Regular $3(s + 5) = 105$ **(or an equivalent equation); $s = 30$. Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

Each student now has $s + 5$, and there are 3 students: $3(s + 5) = 105$.
Divide both sides by 3: $s + 5 = 35$.
Subtract 5 from both sides: $s = 30$.

4. Regular $g = 3\frac{1}{3} = \frac{10}{3}$

Add 4 to both sides: $-1\frac{7}{9} + 4 = 2\frac{2}{9}$, so $2\frac{2}{9} = \left(\frac{2}{3}\right) g$.
Divide both sides by $\frac{2}{3}$, which means multiplying by its reciprocal $\frac{3}{2}$: $g = \frac{20}{9} \times \frac{3}{2} = 3\frac{1}{3}$.

5. Rigor $24.5n - 35 = 185.5$; $n = 9$; **Gym A, by $53.00. Score: 1 point for the equation (or an equivalent equation), $n = 9$ and the correct gym with the amount; n alone earns 0.**

Monthly charges minus the coupon equal the total: $24.5n - 35 = 185.5$.
Add 35 to both sides: $24.5n = 220.5$. Divide both sides by 24.5: $n = 9$.
Gym B for 9 months: $9 \times \$26.50 = \$238.50$.
$\$238.50 - \$185.50 = \$53.00$, so Gym A would have cost $53.00 less.

Re-check strip (3 items, C-AR.2.2-R)

1. Regular $x = -9$

Subtract 4 from both sides: $36 = -4x$.
Divide both sides by −4: $x = -9$.

2. Regular $6(b - 1.25) = 40.5$ **(or an equivalent equation); $b = 8$. Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

Each box without packaging weighs $b - 1.25$, and there are 6 boxes: $6(b - 1.25) = 40.5$.
Divide both sides by 6: $b - 1.25 = 6.75$.
Add 1.25 to both sides: $b = 8$.

3. Regular **A.** $3(x - 8) = 3$

"Decreased by 8" is $(x - 8)$.
Multiplying that result by 3 gives $3(x - 8)$, and that equals 3.

Comparison strip (2 items, C-AR.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $3.75g + 18 = 44.25$; $g = 7$. **Score: 1 point for the equation (or an equivalent equation) and its solution; the solution alone earns 0.**

The cost of the games plus the entry fee equals the total paid: $3.75g + 18 = 44.25$.
Subtract 18 from both sides: $3.75g = 26.25$.
Divide both sides by 3.75: $g = 7$.

2. Regular $t = 2\frac{1}{3} = \frac{7}{3}$

Add 9 to both sides: $-5\frac{8}{9} + 9 = 3\frac{1}{9}$, so $3\frac{1}{9} = \left(\frac{4}{3}\right) t$.
Divide both sides by $\frac{4}{3}$, which means multiplying by its reciprocal $\frac{3}{4}$: $t = \frac{28}{9} \times \frac{3}{4} = 2\frac{1}{3}$.

Reteach plan — C-AR.2.2

MA.7.AR.2.2 · Two-step equations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.2.2-R

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. Solve the equation for x.

$40 = -4x + 4$

2. 6 identical boxes each weigh b pounds with the packaging. After 1.25 pounds of packaging is removed from each box, the 6 boxes weigh 40.5 pounds in all.

Write an equation that can be used to find b. Then solve the equation.

3. A number, x, is decreased by 8. That result is then multiplied by 3, giving 3.

Which equation represents this situation?

A. $3(x - 8) = 3$

B. $3x - 24 = 9$

C. $3(x + 8) = 3$

D. $3x - 8 = 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-AR.2.2-R

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7th Grade Math · Comparison items (2) C-AR.2.2-X

Benchmark: **MA.7.AR.2.2** · Two-step equations with rational numbers Student code: _____ Calculator: yes

1. A park charges a $18.00 entry fee plus $3.75 for each game. A visitor paid $44.25 in all.

Write an equation that can be used to find g, the number of games the visitor paid for. Then solve the equation.

2. What is the exact value of t in the equation shown?

$$-5\frac{8}{9} = \left(\frac{4}{3}\right)t - 9$$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.2.2-X

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